MIL-D-60866(MU)
30 October 1968
SUPERSEDING
FA-PD-MI-2454 Rev 0
27 December 1960

MILITARY SPECIFICATION
DELAY ELEMENT, CARTRIDGE: M27 ASSEMBLY

1. SCOPE

1.1 This specification covers Delay Element, Cartridge: M27 Assembly for use in cartridges for propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specifications For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-13739	- Composition, Delay
MIL-C-60539	- Coating, Anodic, Conventional For Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

FSC 1377

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DRAWINGS

Munitions Command

D10520336	- Delay Element, Cartridge: M27, Assembly
C8826729	- Box, Packing, Ammunition For Delay Elements
C8864162	- Ammunition Container, Metal, M21 For Delay Element, XM23, XM24, XM25, XM27, XM43, XM44 and XM45.
IEL-10520336	- Inspection Equipment List for Delay Element, Cartridge: M27, Assembly.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.- The delay element assembly shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, delay charge or derangement of components.

3.4 Functional

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3.4.1 The delay element shall not fail to fire.

3.4.2 Ballistics.- The delay charge shall consist of increments of type II and type III formulation shown in MIL-C-13739 in proportions and quantities fixed by the contractor to produce the delay time required in table I (see 6.2).

TABLE I

Temperature	Delay Time, Seconds (See 6.3)
-65° to -70°F	6.0 max.
65° to 75°F	5.0 ± 0.5
160° to 165°F	4.0 min.

3.5 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample

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4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 delay bodies, 23 primer retainers and 21 delay element assemblies to be tested as indicated in table II and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

TABLE II

Paragraphs				Type of Test	
Requirement	Test	Test	Sample Size	Dest.	Non-Destr.
3.2	4.4.1	Protective Finish	23 Delay Bodies	X	-
			23 Primer retainers	X	-
3.3	4.4.2	Vibration	12 (from ballistics sample)	-	X
3.4.2	4.4.3.1	Ballistics	21 Delay Elements	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Vibrated	Not Vibrated
-65° to - 70°F	4	3
65° to 75°F	4	3
160° to 165°F	4	3

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- A lot shall consist of an identifiable group of metal parts assemblies produced by a homogeneous process and loaded with accepted delay charges and primers from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defect, using the sampling plan and associated criteria in tables IV and V. For an individual defect (single characteristic) the criteria in table IV shall be used. For total defects within a major or minor class, the criteria in table V shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power magnification (4X) may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE V

Sample Size for Major Defects	Total allowable defects per sample size			
	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

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TABLE V (Cont'd)

Sample Size for Minor Defects	Total Allowable Defects per sample size			
	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Primer, M42, Drawing C8799925, covering a detail of Drawing D-10520336

Defect	Method of Inspection
CRITICAL	
1. Mixed primers	Visual
2. Primer pellet broken, cracked or dislodged	Visual
3. Primer foil discolored	Visual
4. Metal defective	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined.	

4.3.2.2 Body, Delay, Drawing C8594081, covering a detail of Drawing D-10520336

CRITICAL	
1. Diameter of flash hole	Gage
2. Metal defective	Visual
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Minor diameter of thread, min.	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
103.	Concentricity of body with pitch diameter of thread	Gage
104.	Diameter of thread relief	Gage
105.	Diameter of step at thread relief	Gage
106.	Diameter of cavity, max.	Gage
107.	Diameter of body, min.	Gage
108.	Depth of cavity, max.	Gage
109.	Length from cavity face to body diameter	Gage
110.	Thread damaged or not full	Visual
111.	Protective finish inadequate or defective	Visual
112.	Burr	Visual
113.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, max.	Gage
202.	Diameter of cavity, min.	Gage
203.	Diameter of body, max.	Gage
204.	Diameter of step at thread relief	Gage
205.	Diameter of wrench holes	Gage
206.	Depty of cavity, min.	Gage
207.	Depth of wrench holes	Gage
208.	Total length	Gage
209.	Length of thread pilot	Gage
210.	Length of thread relief	Gage
211.	Length of step at thread relief	Gage
212.	Location of wrench holes	Gage
213.	Radius or chamfer missing or incorrect	Visual
214.	Surface finish improper	Visual

4.3.2.3 Gasket, Aluminum, Drawing B8594082, covering a detail of Drawing D-10520336

CRITICAL

None defined

MAJOR

101.	Diameter inside, max.	Gage
102.	Metal defective	Visual
103.	Burr	Visual
104.	Foreign matter	Visual

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter inside, min.	Gage
202. Diameter outside, max.	Gage
203. Total thickness	Gage
204. Surface finish improper	Visual

4.3.2.4 Cup, Retaining, Drawing B8597232, covering a detail of Drawing D-10520336

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of large cavity with small cavity	Gage
102. Concentricity of body with small cavity	Gage
103. Diameter of large cavity, max.	Gage
104. Diameter of small cavity, max.	Gage
105. Diameter of body, min.	Gage
106. Depth of large cavity	Gage
107. Surface finish of small cavity with body	Visual
108. Burr or sharp corner	Visual
109. Foreign matter	Visual
MINOR	
201. Diameter of large cavity, min.	Gage
202. Diameter of small cavity, min.	Gage
203. Diameter of body, max.	Gage
204. Total length	Gage
205. Surface finish improper, except as otherwise classified	Visual

4.3.2.5 Plug, Firing, Drawing B8597234, covering a detail of Drawing D-10520336.

CRITICAL	
1. Metal defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of flange with body	Gage
102. Diameter of flange, min.	Gage
103. Diameter of body, min.	Gage
104. Length of striker	Gage
105. Radius of striker	Gage
106. Surface finish of body improper	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter of flange, max.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Location of flange	Gage
205. Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Retainer, Primer, Drawing C10520003, covering a detail of Drawing D-10520336.

CRITICAL	
1. Depth of head counterbore	Gage
2. Depth of primer pocket	Gage
3. Depth of cup pocket	Gage
4. Metal defective	Visual
MAJOR	
101. Pitch diameter of internal thread, max.	Gage
102. Minor diameter of internal thread, max.	Gage
103. Concentricity of flange with the primer pocket	Gage
104. Concentricity of flash hole with primer pocket	Gage
105. Concentricity of large end with primer pocket	Gage

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4.3.2.6 (Cont'd)

	Defect	Method of Inspection
106.	Concentricity of head counterbore with primer pocket	Gage
107.	Concentricity of cup pocket with primer pocket	Gage
108.	Concentricity of body with primer pocket	Gage
109.	Concentricity of pitch diameter of thread with body	Gage
110.	Symmetry of wrench flats with pitch diameter of thread	Gage
111.	Diameter of body, min.	Gage
112.	Diameter of primer pocket, min.	Gage
113.	Diameter of cup pocket, min.	Gage
114.	Diameter of head counterbore	Gage
115.	Depth of primer pocket	Gage
116.	Depth of cup pocket	Gage
117.	Length of effective thread, min.	Gage
118.	Location of flange	Gage
119.	Location of datum diameter	Gage
120.	Thread damaged or not full	Visual
121.	Protective finish inadequate or defective	Visual
122.	Surface finish of primer pocket improper	Visual
123.	Burr or sharp corner	Visual
124.	Foreign matter	Visual

MINOR

201.	Pitch diameter of internal thread, min.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of primer pocket, max.	Gage
204.	Diameter of cup pocket, max.	Gage
205.	Diameter of head	Gage
206.	Depth of wrench flats	Gage
207.	Depth to bottom of threaded cavity	Gage
208.	Length of wrench flats	Gage
209.	Length across wrench flats	Gage
210.	Total length	Gage
211.	Location of wrench flats	Gage
212.	Radius or chamfer missing or incorrect	Visual
213.	Surface finish improper, except as otherwise classified	Visual

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4.3.2.7 Assembly, Delay Element, Cartridge, M27, Drawing D-10520336

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray 1/ and visual
2. Improper assembly	X-ray 1/ and visual
3. Primer not properly seated	X-ray 1/ and visual
4. Mixed ammunition	Visual
5. Flash hole missing or obstructed	Visual
6. Primer uncrimped	Visual
7. Marking missing, incorrect or illegible	Visual
8. Delay charge not visible through flash hole	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Diameter of body, max.	Gage
103. Retainer cup not properly seated	Visual
104. Firing plug not flush or below flush with retaining cup	Visual
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Lacquer missing or inadequate	Visual

1/ Prior to assembly into a cartridge the delay element shall be x-rayed with the side of the case perpendicular to the rays of the X-ray machine. The x-ray negatives properly identified, shall be forwarded to the procuring activity.

4.3.2.8 Ammunition Container, Drawing C8864162.

CRITICAL

None defined.

MAJOR

None defined.

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Sealing inadequate, incomplete or incorrect	Visual
205. Container cut or damaged	Visual
206. Marking missing, incorrect, or illegible	Visual

4.3.2.9 Box, Packing, Drawing C8826729

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Content of box exposed	Visual
205. Box damaged	Visual
206. Contents incorrect	Visual
207. Marking missing, incorrect or illegible	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- Twenty-three delay bodies and 23 primer retainers shall be selected from each anodized lot for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot of components.

4.3.3.2 Vibration.- Twelve of the 21 assembled x-rayed delay elements of the initial production sample shall be subjected to vibration in accordance with 4.4.2. After vibration, the 12 vibrated delay elements shall be X-rayed

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as directed in note 1/ 4.3.2.7. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the delay element or in the lot of primers, 12 additional delay elements shall be selected and tested as indicated herein.

4.3.3.3 Functional.- The following tests shall be performed by the Government:

4.3.3.3.1 Ballistics.- Twenty-one test delay elements from each production lot shall be grouped and temperature conditioned in accordance with 4.4.3.1.1 and table VI. Failure of any delay element to comply with the requirements of 3.4 shall be cause for rejection of the lot.

TABLE VI

Temperature	Sample Size
-65° to -70°F	7
65° to 75°F	7
160° to 165°F	7

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 10520336.

4.4 Test methods and procedures

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539. The delay body and primer retainer shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Functional.

4.4.3.1 Ballistics.- Delay elements of the sample shall be grouped and temperature conditioned in accordance with 4.4.3.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be re-conditioned in accordance with 4.4.3.1.1.

4.4.3.1.1 Temperature conditioning.- The vent holes of the delay element shall be sealed against environmental atmosphere of the conditioning chamber. A standard inert delay element, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1 hour longer than the time

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necessary to bring the element contained in the dummy delay element to the required temperature. Upon completion of the temperature conditioning, the vent seal shall be removed.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawing C8826729 and C8864162, except that all markings shall refer to Delay Element, Cartridge M27, Assembly.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTE

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 Advisory delay charge composition (Formulation).- consists of 3 increments of charge as follows: approximately 5 grains type II, 13½ grains type III and 5 grains type II, charged in that order.

6.3 Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the vent hole of the delay element.

6.4 Supersession date.- This specification includes the requirements of Purchase Description FA-PD-MI-2454 dated 27 December 1960.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A231

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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